

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2021 P1H Worked Solutions

Jan 2021 | P1H

29 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Ratio Problem Solving**

Jan 2021 P1H - Jan 2021 | P1H

- 1** Pieter owns a currency conversion shop.

Last Monday, Pieter changed a total of 20 160 rand into a number of different currencies.

He changed $\frac{3}{10}$ of the 20 160 rand into euros.

He changed the rest of the rands into dollars, rupees and francs in the ratios 9 : 5 : 2

Pieter changed more rands into dollars than he changed into francs.

Work out how many more.

..... rand

(Total for Question 1 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 1

Marks: 4

TOPIC: **Ratio Problem Solving**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 5****TOPIC: Statistics Toolkit**

Jan 2021 P1H - Jan 2021 | P1H

- 2 The table gives information about the speeds, in kilometres per hour, of 80 motorbikes as each pass under a bridge.

Speed (s kilometres per hour)	Frequency
$40 < s \leq 50$	10
$50 < s \leq 60$	16
$60 < s \leq 70$	19
$70 < s \leq 80$	23
$80 < s \leq 90$	12

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean speed of the motorbikes as they pass under the bridge.
Give your answer correct to 3 significant figures.

..... kilometres per hour

(4)

(Total for Question 2 is 5 marks)

PAST PAPER SOLUTION

Q#: 2

Marks: 5

TOPIC: **Statistics Toolkit**

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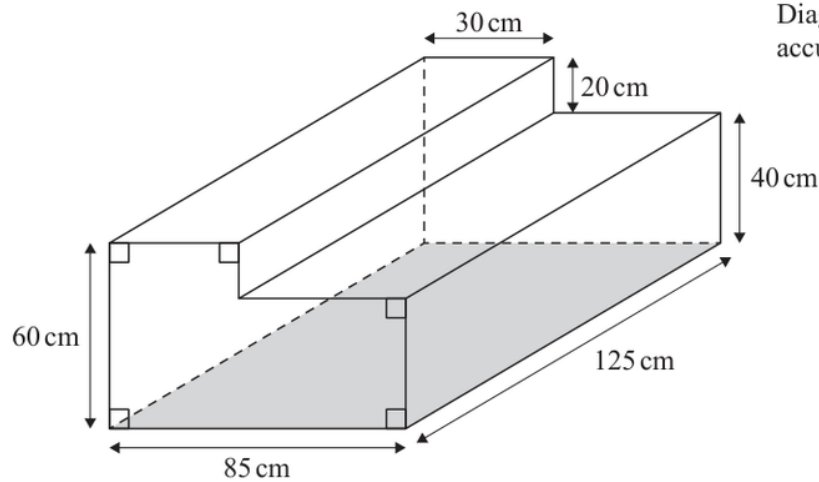
No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 4****TOPIC: Volume & Surface Area**

Jan 2021 P1H - Jan 2021 | P1H

- 3 The diagram shows a container for water in the shape of a prism.



The rectangular base of the prism, shown shaded in the diagram, is horizontal.
The container is completely full of water.

Tuah is going to use a pump to empty the water from the container so that the volume of water in the container decreases at a constant rate.

The pump starts to empty water from the container at 10 30 and at 12 00 the water level in the container has dropped by 20 cm.

Find the time at which all the water has been pumped out of the container.

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3

Marks: 4

TOPIC: **Volume & Surface Area**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 2****TOPIC: Set Notation & Venn Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

4 $\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$

$$A = \{\text{odd numbers}\}$$

$$B = \{\text{multiples of 3}\}$$

List the members of the set

(i) $A \cap B$

.....
(1)

(ii) $A \cup B$

.....
(1)

.....
(Total for Question 4 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 4

Marks: 2

TOPIC: **Set Notation & Venn Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 5****Marks: 5****TOPIC: Solving Linear Equations**

Jan 2021 P1H - Jan 2021 | P1H

5 (a) Factorise fully $15y^4 + 20uy^3$
(2)(b) Solve $4 - 3x = \frac{5 - 8x}{4}$

Show clear algebraic working.

 $x =$
(3)**(Total for Question 5 is 5 marks)**

EA

PAST PAPER SOLUTION

Q#: 5

Marks: 5

TOPIC: **Solving Linear Equations**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 2****TOPIC: Powers, Roots & Standard Form**

Jan 2021 P1H - Jan 2021 | P1H

6 (a) Write 2840000000 in standard form.

.....
(1)

(b) Write 2.5×10^{-4} as an ordinary number.

.....
(1)

(Total for Question 6 is 2 marks)

EA

PAST PAPER SOLUTION

Q#: 6

Marks: 2

TOPIC: **Powers, Roots & Standard Form**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 7****Marks: 6****TOPIC: Compound Interest & Depreciation**

Jan 2021 P1H - Jan 2021 | P1H

- 7 Chen invests 40 000 yuan in a fixed-term bond for 3 years.

The fixed-term bond pays compound interest at a rate of 3.5% each year.

- (a) Work out the value of Chen's investment at the end of 3 years.
Give your answer to the nearest yuan.

..... yuan
(3)

Wang invested P yuan.

The value of his investment decreased by 6.5% each year.

At the end of the first year, the value of Wang's investment was 30 481 yuan.

- (b) Work out the value of P .

$P =$
(3)

(Total for Question 7 is 6 marks)

PAST PAPER SOLUTION

Q#: 7

Marks: 6

TOPIC: **Compound Interest & Depreciation**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P1H - Jan 2021 | P1H

- 8 The region, shown shaded in the diagram, is a path.

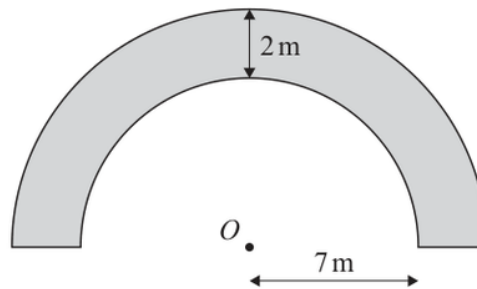


Diagram **NOT**
accurately drawn

The boundary of the path is formed by two semicircles, with the same centre O , and two straight lines.

The inner semicircle has a radius of 7 metres.

The path has a width of 2 metres.

Work out the perimeter of the path.

Give your answer correct to one decimal place.

..... m

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8

Marks: 3

TOPIC: **Circles, Arcs & Sectors**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 9****Marks: 5****TOPIC: Factorising**

Jan 2021 P1H - Jan 2021 | P1H

9 (a) Simplify $(2x^3y^5)^4$
(2)(b) (i) Factorise $x^2 + 5x - 36$
(2)(ii) Hence, solve $x^2 + 5x - 36 = 0$
(1)

(Total for Question 9 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 9

Marks: 5

TOPIC: **Factorising**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

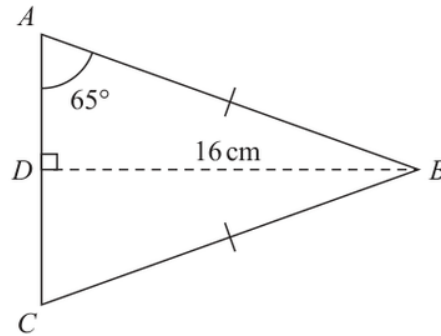
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EA

PAST PAPER QUESTION**Q#: 10****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2021 P1H - Jan 2021 | P1H

10 Here is isosceles triangle ABC .Diagram **NOT**
accurately drawn D is the midpoint of AC and $DB = 16$ cm.Angle $DAB = 65^\circ$ Work out the perimeter of triangle ABC .

Give your answer correct to one decimal place.

..... cm

(Total for Question 10 is 4 marks)

PAST PAPER SOLUTION

Q#: 10

Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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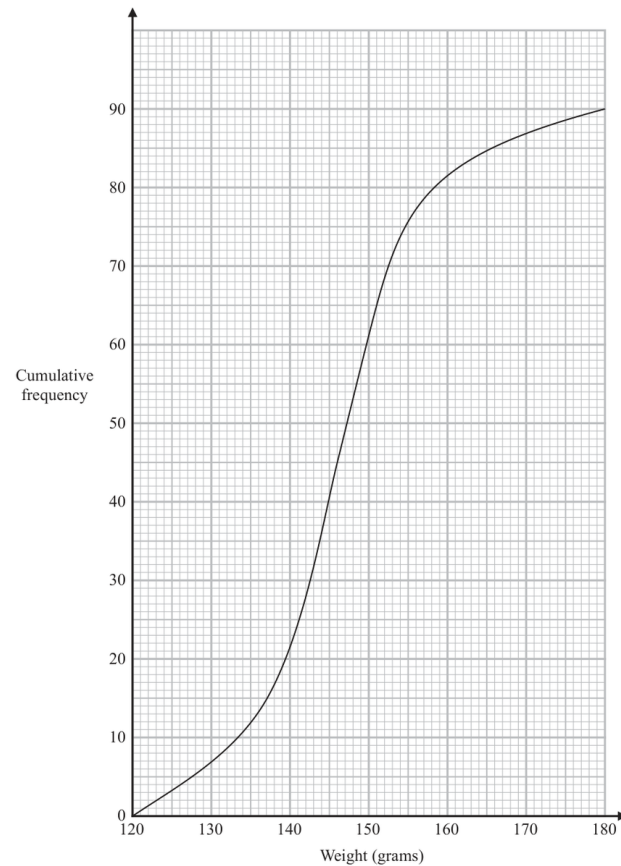
No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 11****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

- 11 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets.



- (a) Find an estimate for the median of the weights of these bags of sweets.

..... grams
(2)

Roberto sells the bags of sweets to raise money for charity.
Bags with a weight greater than d grams are labelled large bags and sold for 3.75 euros each bag.

The total amount of money he receives by selling all the large bags is 93.75 euros.

- (b) Find the value of d .

$d =$
(3)

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

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WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 12****Marks: 6****TOPIC: Expanding Brackets**

Jan 2021 P1H - Jan 2021 | P1H

12 (a) Express $\frac{4}{x-2} - \frac{3}{x+1}$ as a single fraction.

Give your answer in its simplest form.

.....
(3)

(b) Expand and simplify $2x(x-5)(x-3)$

.....
(3)

(Total for Question 12 is 6 marks)

EA

PAST PAPER SOLUTION

Q#: 12

Marks: 6

TOPIC: **Expanding Brackets**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Coordinate Geometry**

Jan 2021 P1H - Jan 2021 | P1H

- 13** Point A has coordinates $(5, 8)$
Point B has coordinates $(9, -4)$

(a) Work out the gradient of AB .

.....
(2)

The straight line L has equation $y = -4x + 5$

(b) Write down the gradient of a straight line that is perpendicular to L .

.....
(1)

(Total for Question 13 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 13

Marks: 3

TOPIC: **Coordinate Geometry**

Jan 2021 P1H - Jan 2021 | P1H

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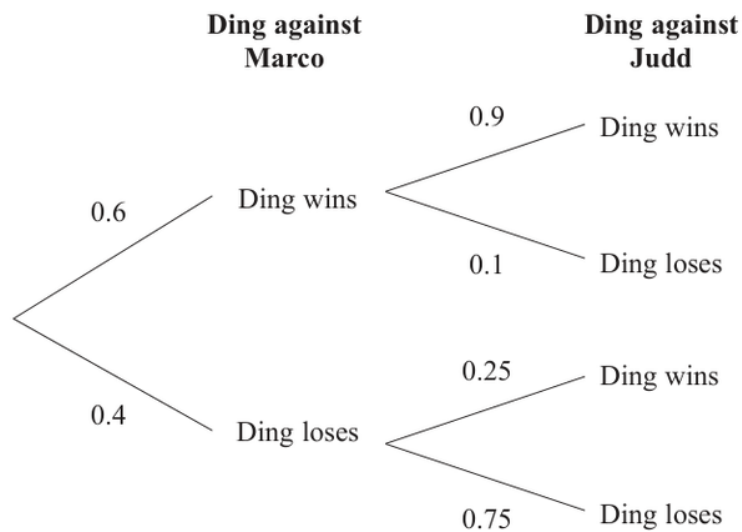
EA

PAST PAPER QUESTION**Q#: 14****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

- 14** Ding is going to play one game of snooker against each of two of his friends, Marco and Judd.

The probability tree diagram gives information about the probabilities that Ding will win or lose each of these two games.



- (a) Work out the probability that Ding will win both games.

.....
(2)

- (b) Work out the probability that Ding will win exactly one of the games.

.....
(3)

(Total for Question 14 is 5 marks)

PAST PAPER SOLUTION

Q#: 14

Marks: 5

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2021 P1H - Jan 2021 | P1H

15

$$a = \frac{v - u}{t}$$

 $v = 9.6$ correct to 1 decimal place $u = 3.8$ correct to 1 decimal place $t = 1.84$ correct to 2 decimal placesCalculate the upper bound for the value of a .

Give your answer as a decimal correct to 2 decimal places.

Show your working clearly.

(Total for Question 15 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 15

Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

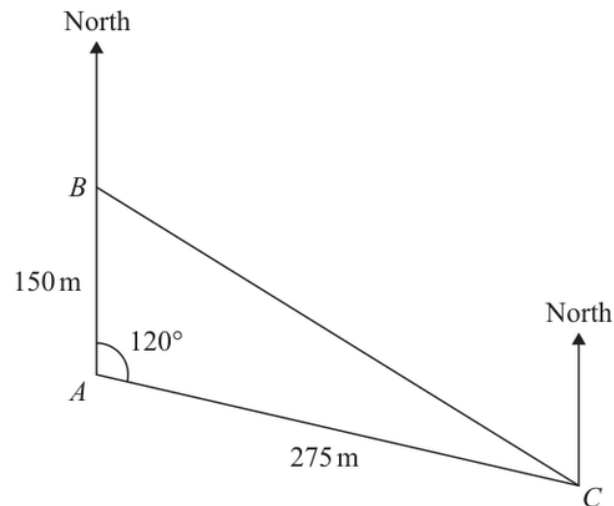
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 16****Marks: 5****TOPIC: Bearings, Scale Drawing & Constructions**

Jan 2021 P1H - Jan 2021 | P1H

16 The diagram shows the positions of three ships, A , B and C .Diagram **NOT**
accurately drawnShip B is due north of ship A .The bearing of ship C from ship A is 120° .Calculate the bearing of ship C from ship B .

Give your answer correct to the nearest degree.

o

(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION

Q#: 16

Marks: 5

TOPIC: **Bearings, Scale Drawing & Constructions**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 17****Marks: 5****TOPIC: Differentiation**

Jan 2021 P1H - Jan 2021 | P1H

17 A solid, **S**, is made from a hemisphere and a cylinder.

The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point.

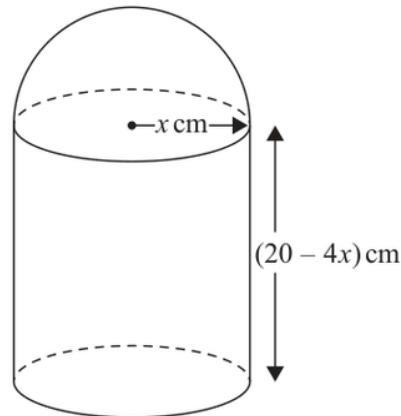


Diagram **NOT**
accurately drawn

The radius of the cylinder and the radius of the hemisphere are both x cm.

The height of the cylinder is $(20 - 4x)$ cm.

The volume of **S** is $V \text{ cm}^3$ where $V = \frac{1}{3} \pi y$

Find the maximum value of y .

Show clear algebraic working.

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION

Q#: 17

Marks: 5

TOPIC: **Differentiation**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 3**TOPIC: **Surds**

Jan 2021 P1H - Jan 2021 | P1H

- 18** Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is a prime number and y is an integer, find the value of x and the value of y . Show each stage of your working clearly.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$ **(Total for Question 18 is 3 marks)**

EA

PAST PAPER SOLUTION

Q#: 18

Marks: 3

TOPIC: **Surds**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Simultaneous Equations**

Jan 2021 P1H - Jan 2021 | P1H

19 Solve the simultaneous equations

$$x^2 - 9y - x = 2y^2 - 12$$

$$x + 2y - 1 = 0$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 19

Marks: 5

TOPIC: **Simultaneous Equations**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

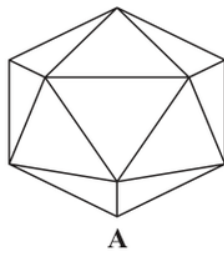
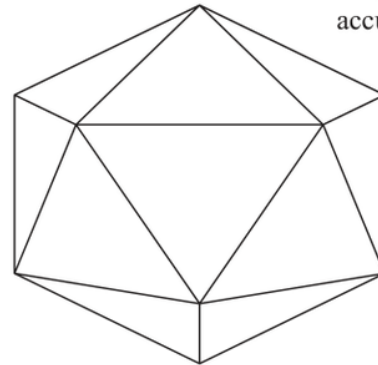
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

20 **A** and **B** are two similar solids.**A****B**Diagram **NOT**
accurately drawn**A** has a volume of 1836 cm^3 **B** has a volume of 4352 cm^3 **B** has a total surface area of 1120 cm^2 Work out the total surface area of **A**...... cm^2

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

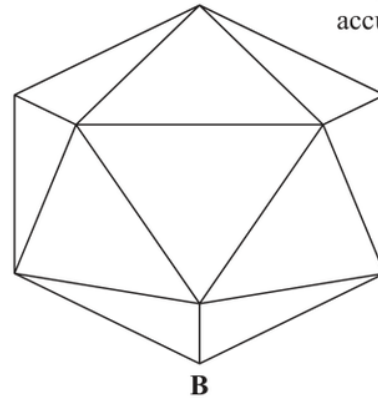
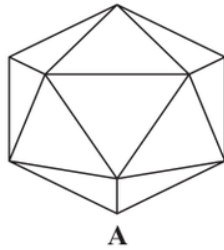
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

20 **A** and **B** are two similar solids.Diagram **NOT**
accurately drawn**A** has a volume of 1836 cm^3 **B** has a volume of 4352 cm^3 **B** has a total surface area of 1120 cm^2 Work out the total surface area of **A**...... cm^2

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

21 A curve has equation $y = f(x)$ The coordinates of the minimum point on this curve are $(-9, 15)$

(a) Write down the coordinates of the minimum point on the curve with equation

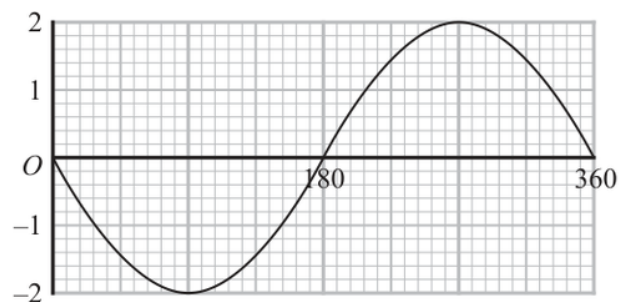
(i) $y = f(x + 3)$

(.....,)

(ii) $y = \frac{1}{3}f(x)$

(.....,)

(2)

The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.Given that $a > 0$ and that $0 < b < 360$ (b) find the value of a and the value of b . $a = \dots\dots\dots$ $b = \dots\dots\dots$

(2)

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

21 A curve has equation $y = f(x)$ The coordinates of the minimum point on this curve are $(-9, 15)$

(a) Write down the coordinates of the minimum point on the curve with equation

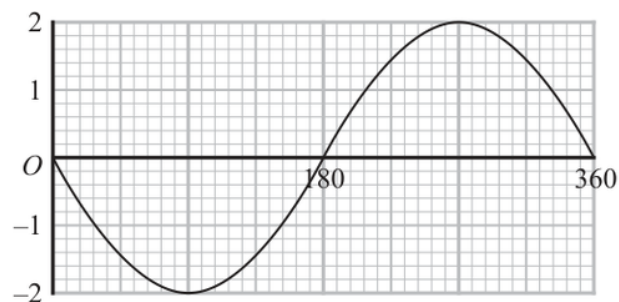
(i) $y = f(x + 3)$

(.....,)

(ii) $y = \frac{1}{3}f(x)$

(.....,)

(2)

The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.Given that $a > 0$ and that $0 < b < 360$ (b) find the value of a and the value of b . $a = \dots\dots\dots$ $b = \dots\dots\dots$

(2)

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 4

TOPIC: **Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 3****TOPIC: Functions**

Jan 2021 P1H - Jan 2021 | P1H

22 The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leq 4$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 22 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 3

TOPIC: **Functions**

Jan 2021 P1H - Jan 2021 | P1H

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PAST PAPER QUESTION**Q#: 22****Marks: 3****TOPIC: Functions**

Jan 2021 P1H - Jan 2021 | P1H

22 The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leq 4$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 22 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 3

TOPIC: **Functions**

Jan 2021 P1H - Jan 2021 | P1H

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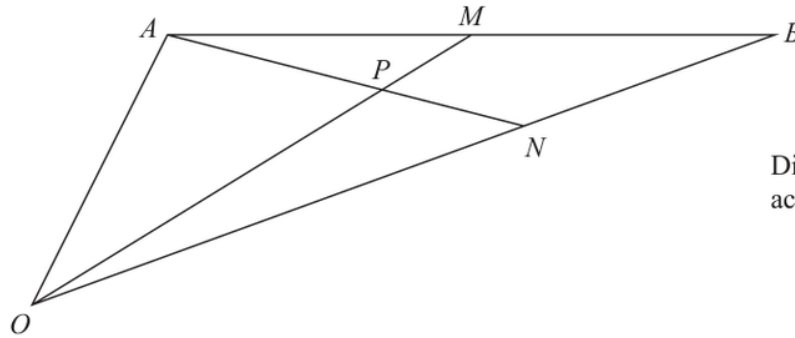
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 6**TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

23 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \text{ and } \vec{OB} = 2\mathbf{b}$$

 M is the midpoint of AB . N is the point on OB such that $ON:NB = 2:1$ P is the point on AN such that OPM is a straight line.Use a vector method to find $OP:PM$

Show your working clearly.

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 6

TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

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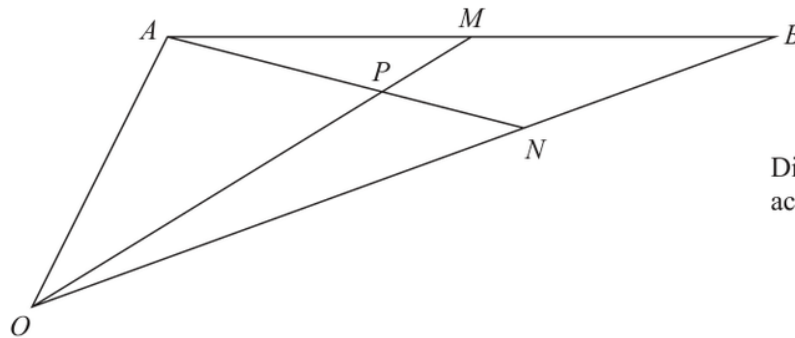
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EA

PAST PAPER QUESTION**Q#: 23****Marks: 6**TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

23 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \text{ and } \vec{OB} = 2\mathbf{b}$$

 M is the midpoint of AB . N is the point on OB such that $ON:NB = 2:1$ P is the point on AN such that OPM is a straight line.Use a vector method to find $OP:PM$

Show your working clearly.

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 6

TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Sequences**

Jan 2021 P1H - Jan 2021 | P1H

24 An arithmetic series has first term a and common difference d .

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d .
Show your working clearly.

 $a = \dots\dots\dots$

(Total for Question 24 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 24

Marks: 4

TOPIC: **Sequences**

Jan 2021 P1H - Jan 2021 | P1H

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PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Sequences**

Jan 2021 P1H - Jan 2021 | P1H

24 An arithmetic series has first term a and common difference d .

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d .
Show your working clearly.

 $a = \dots\dots\dots$

(Total for Question 24 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 24

Marks: 4

TOPIC: **Sequences**

Jan 2021 P1H - Jan 2021 | P1H

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